ALGEBRA 1 - UNIT 1

Section 1.1
Multiple Representations of Relationships

Seeing the same relationship in more than one way.

> **Mathematical idea:** A relationship can look different in a table, graph, equation, or story, but the pattern underneath can still be the same.

Today we will practice connecting representations, explaining what changes, and using evidence to decide whether two representations match.

Learning Target

Students will connect tables, graphs, equations, and contexts that represent the same relationship.

I can statements

- I can identify when different representations describe the same relationship.
- I can match tables, graphs, equations, and situations.
- I can describe how two quantities are related.
- I can determine what quantities are changing.
- I can justify how representations are connected.

What’s To Come

This is a long-range preview for future lessons. Do not solve it today. Instead, annotate it individually. Circle anything familiar, underline symbols or directions that seem important, and write notes about what information you think will be needed later.

As you annotate, ask yourself: What quantities are changing? How can the table, graph, equation, and context show the same pattern? What evidence would prove that two representations match? What information would make a representation not match?

Future Task

A school walkathon raises a starting donation plus the same amount for each mile walked. Three students claim they have representations for the same relationship.

Context

The team already has $10 in donations. They earn $5 more for each mile walked.

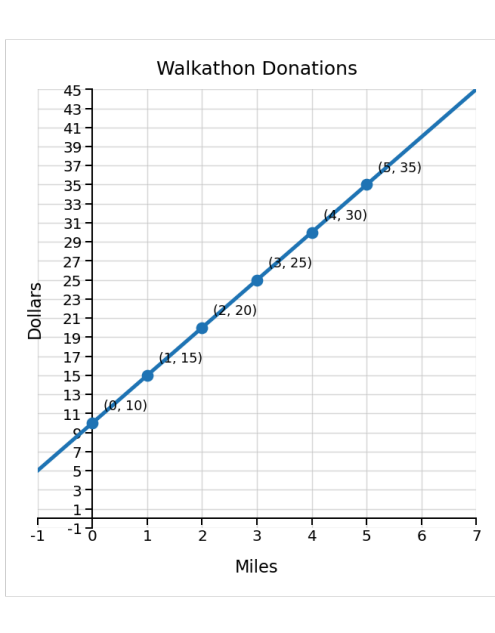
Equation

$d = 5m + 10$, where m is miles and d is dollars.

Table

Miles	0	1	2	3
Dollars	10	15	20	25

Graph



1. Use at least three pieces of evidence to justify whether the representations match.
2. Explain what the starting donation means in each representation.
3. Explain what the amount per mile means in each representation.
4. A classmate says the table must be wrong because the miles do not start at 1. Write a response that corrects the misunderstanding.

Intro

Directions

Read the introduction aloud with a partner or in a small group. As you read, annotate the text by marking important ideas, highlighting or circling key vocabulary, and writing short notes about connections, questions, or predictions in the margins.

A relationship tells how two things are connected. In math, the two things are often quantities that can change. For example, the number of rides on a bus and the total cost can be connected.

We can show the same relationship in different ways. A table shows organized input and output values. A graph shows the pattern as points or a line. An equation uses symbols to describe the rule. A context tells the story in words.

The important question is not, “Do these look the same?” The important question is, “Do these show the same relationship?” To answer that, we look for evidence. We check what starts the pattern, what changes each time, and what the numbers mean.

When you connect representations, you are building a stronger picture of the relationship. One representation may be easier to read than another, but they should all agree about how the two quantities are related.

Stop and Discuss

1. What is a relationship in math?
2. How can a table, graph, equation, and context show the same idea?
3. What evidence could help you decide whether two representations match?

One Relationship, Four Ways

Each representation can tell part of the story. When they match, the numbers and the meaning should agree.

Input: the starting value you choose or measure.

Output: the value that depends on the input.

Relationship: the way the input and output are connected.

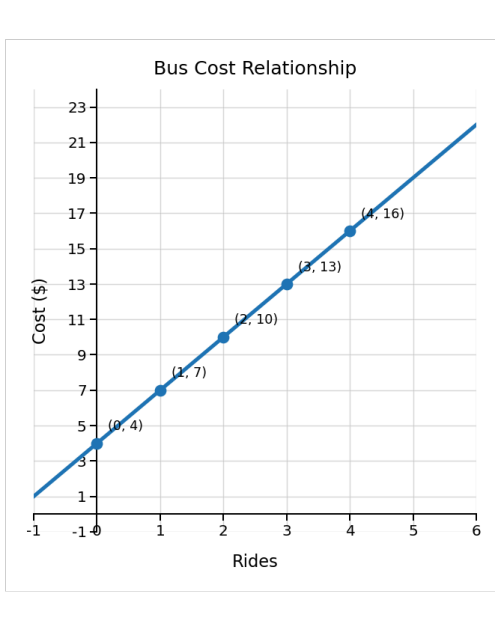
Example 1

A bus pass costs $4 to buy. Each ride costs $3. The total cost is represented by $c = 3r + 4$.

Table

Rides, r	0	1	2	3
Cost, c	4	7	10	13

Graph



Explain how the context, equation, table, and graph each show the starting cost and the cost per ride.

You Try It 1

A gym charges a one-time membership fee of $15 and then $2 for each visit.

Write an equation for the total cost, make a quick table for 0, 1, 2, and 3 visits, and explain where the starting fee and cost per visit would appear on a graph.

Stop and Discuss

1. Where do you see the 4-dollar starting cost in the table, graph, equation, and context?
2. Where do you see the 3-dollar-per-ride change in the table, graph, equation, and context?

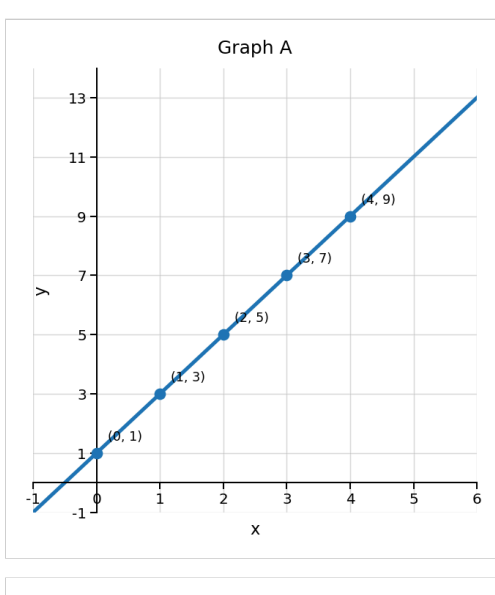
Matching by What Changes

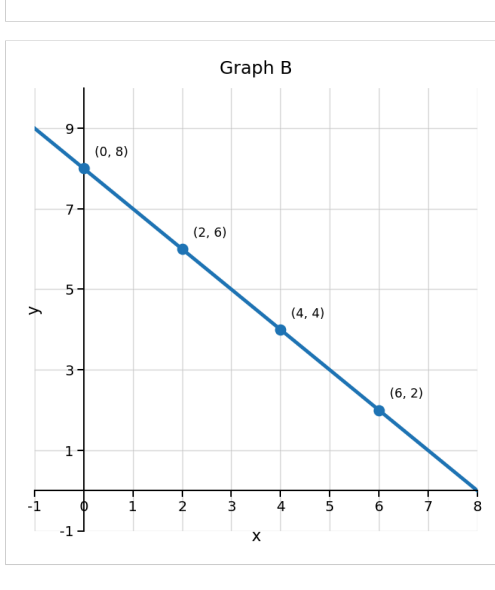
Two representations can match even if they look very different. Look for the same input-output pairs, the same starting value, and the same change.

Example 2

Which graph matches the table? Use evidence from the table and the graphs.

x	0	1	2	3	4
y	1	3	5	7	9





Write a sentence that uses at least two table values as evidence.

You Try It 2

A different table starts with $y = 8$ when $x = 0$ and then decreases by 1 each time x increases by 1.

Which graph above would match that table? Explain using at least two pieces of evidence.

Stop and Discuss

1. What evidence should you check first when matching a table to a graph?
2. Why is one point usually not enough evidence to prove a match?

Reading Each Representation for Meaning

A representation is more useful when we can explain what the numbers mean. We should connect numbers to the situation, not just copy them.

Example 3

A water tank already has 12 gallons in it. Water is added at a steady rate of 4 gallons each minute.

1. Choose the equation that matches: $w = 12m + 4$ or $w = 4m + 12$.
2. Explain what the input represents.
3. Explain what the output represents.
4. Describe how the table should change from one row to the next.

You Try It 3

A phone battery already has 30% charge. It gains 5% each minute while charging.

1. Write an equation that matches the situation.
2. Explain what the input and output represent.
3. Describe how the table should change from one row to the next.

Stop and Discuss

1. Why is $w = 12m + 4$ a tempting mistake?
2. How can the words “already has” help you choose the equation?

When Representations Do Not Match

Sometimes one representation has an error. To find it, compare the meaning, not just the numbers.

Example 4

A school club sells shirts. The club pays a $20 setup fee and then pays $6 for each shirt.

A student writes the equation $c = 20s + 6$.

1. Explain the student’s mistake.
2. Write the correct equation.
3. Describe what a matching table would show when $s = 0$, $s = 1$, and $s = 2$.
4. Explain how a graph would show the setup fee.

You Try It 4

A fundraiser has a $12 starting donation and earns $4 for each bracelet sold. A student writes $d = 12b + 4$.

1. Explain the mistake.
2. Write the correct equation.
3. Describe one table value and one graph feature that would prove your equation matches the situation.

Stop and Discuss

1. How can an error in an equation create an error in the table and graph?
2. What evidence would help you catch the error before graphing?

Summary

- A relationship connects two quantities.
- Tables, graphs, equations, and contexts can show the same relationship.
- A table shows input-output pairs.
- A graph shows the pattern visually.
- An equation gives a rule using symbols.
- A context explains what the quantities mean in words.
- Good justification uses evidence from more than one representation.

Common Mistakes

- **Matching by looks only:** Two representations match because the relationship is the same, not because the pages look similar.
- **Mixing up input and output:** Always ask which quantity is chosen or measured first and which quantity depends on it.
- **Ignoring the start value:** The value when the input is 0 can be important evidence.
- **Ignoring the change:** Check how the output changes when the input increases.
- **Swapping numbers in an equation:** In $y = mx + b$, the change rate and starting value have different meanings.
- **Not explaining evidence:** A correct match is stronger when you can say why it matches.

Optional Future Task Reflection

Optional Reflection: Look back at the future task. Do not solve it yet.

1. Highlight one part that makes more sense now than it did at the beginning.
2. Circle one part that still feels unfamiliar.
3. Write one sentence about what representation you would look at first.